

Proposition: Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be distinct eigenvalues of the linear transformation T .

Let $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_k$ be eigenvectors corresponding to $\lambda_1, \lambda_2, \dots, \lambda_k$ respectively.

Then $\{\vec{x}_1, \dots, \vec{x}_k\}$ is linearly independent.

Proof: Let $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_i\} \subseteq \{\vec{x}_1, \dots, \vec{x}_k\}$.

Proof by induction on i .

Base case: when $i=1$ $\{\vec{x}_1\} \neq \{\vec{0}\}$, thus $\{\vec{x}_1\}$ is linearly independent.

Inductive steps. Assume $\{\vec{x}_1, \dots, \vec{x}_i\}$ is linearly independent

$$\text{Suppose } a_1 \vec{x}_1 + a_2 \vec{x}_2 + \dots + a_{i+1} \vec{x}_{i+1} = \vec{0} \quad (*)$$

$$\text{Then } T(a_1 \vec{x}_1 + a_2 \vec{x}_2 + \dots + a_{i+1} \vec{x}_{i+1}) = \vec{0}$$

$$\Leftrightarrow \lambda_1 a_1 \vec{x}_1 + \lambda_2 a_2 \vec{x}_2 + \dots + \lambda_{i+1} a_{i+1} \vec{x}_{i+1} = \vec{0} \quad \textcircled{1}$$

$$\text{multiply } \lambda_{i+1} \text{ to } (*): a_1 \lambda_{i+1} \vec{x}_1 + a_2 \lambda_{i+1} \vec{x}_2 + \dots + a_i \lambda_{i+1} \vec{x}_i + a_{i+1} \lambda_{i+1} \vec{x}_{i+1} = \vec{0} \quad \textcircled{2}$$

$$\textcircled{2} - \textcircled{1}: a_1 (\lambda_{i+1} - \lambda_1) \vec{x}_1 + a_2 (\lambda_{i+1} - \lambda_2) \vec{x}_2 + \dots + a_i (\lambda_{i+1} - \lambda_i) \vec{x}_i = \vec{0}$$

By inductive assumption that $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_i\}$ is linearly independent

$$\Rightarrow a_1 (\lambda_{i+1} - \lambda_1) = a_2 (\lambda_{i+1} - \lambda_2) = \dots = a_i (\lambda_{i+1} - \lambda_i) = 0$$

As $\lambda_1, \dots, \lambda_{i+1}$ are distinct eigenvalues: $\lambda_{i+1} - \lambda_1 \neq 0, \lambda_{i+1} - \lambda_2 \neq 0, \dots, \lambda_{i+1} - \lambda_i \neq 0$

$$\Rightarrow a_1 = a_2 = \dots = a_i = 0$$

$$\text{plug } a_1 = a_2 = \dots = a_i = 0 \text{ into } (*): a_{i+1} \vec{x}_{i+1} = \vec{0}$$

$$\text{Since } \vec{x}_{i+1} \neq \vec{0}, a_{i+1} = 0$$

$$\Rightarrow \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_{i+1}\} \text{ is linearly independent.} \quad // \text{ end of Feb 5}$$

Corollary. Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be distinct eigenvalues of the linear transformation T .

Let $\vec{x}_1 \in E_{\lambda_1}, \vec{x}_2 \in E_{\lambda_2}, \dots, \vec{x}_k \in E_{\lambda_k}$. If $\vec{x}_1 + \vec{x}_2 + \dots + \vec{x}_k = \vec{0}$,

then $\vec{x}_1 = \vec{x}_2 = \dots = \vec{x}_k = \vec{0}$.

Proof Suppose towards a contradiction that there are some non-zero vectors in $\{\vec{x}_1, \dots, \vec{x}_k\}$. Let $S =$ the set of nonzero vectors in $\{\vec{x}_1, \dots, \vec{x}_k\}$.

Without loss of generality, we may assume $S = \{\vec{x}_1, \dots, \vec{x}_i\}$, for some $i \leq k$.

Then $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_i$ are eigenvectors corresponding to $\lambda_1, \dots, \lambda_i$.

Then $\vec{x}_1 + \vec{x}_2 + \dots + \vec{x}_i = \vec{0} \Rightarrow \{\vec{x}_1, \dots, \vec{x}_i\}$ is linearly dependent $\Rightarrow \in$. □

Definition. Let $W_1, W_2, \dots, W_k$ be subspace of V . Define the sum of $W_1, W_2, \dots, W_k$ by:

$$W_1 + W_2 + \dots + W_k = \{ \vec{x}_1 + \vec{x}_2 + \dots + \vec{x}_k : \forall \vec{x}_1 \in W_1, \vec{x}_2 \in W_2, \dots, \vec{x}_k \in W_k \}.$$

The sum $W_1 + W_2 + \dots + W_k$ is called a direct sum, and denoted by $W_1 \oplus W_2 \oplus \dots \oplus W_k$.

$$W_i \cap (W_1 + W_2 + \dots + W_i) = \{\vec{0}\} \text{ for all } i=1, 2, \dots, k-1.$$

Remarks: 1) The idea of defining direct sum is such that for any

$y \in W_1 + \dots + W_k$, there is a unique "decomposition" of $y = \vec{x}_1 + \dots + \vec{x}_k$ with $x_1 \in W_1, x_2 \in W_2, \dots, x_k \in W_k$.

With the above condition: $\forall y \in W_1 \oplus W_2 \oplus \dots \oplus W_k$

Start a unique decomposition of $y = \vec{x}_k + \vec{x}$ for $\vec{x}_k \in W_k$ and $\vec{x} \in W_1 + W_2 + \dots + W_{k-1}$.

In fact if there exists $\vec{x}'_k \in W_k$ and $\vec{x}' \in W_1 + W_2 + \dots + W_{k-1}$.

$$\text{Then } y = \vec{x}_k + \vec{x} = \vec{x}'_k + \vec{x}' \Rightarrow \vec{x}_k - \vec{x}'_k = \vec{x} - \vec{x}'$$

Where $\vec{x}_k - \vec{x}'_k \in W_k$ and $\vec{x} - \vec{x}' \in W_1 + \dots + W_{k-1}$

$$\Rightarrow \vec{x}_k - \vec{x}'_k = \vec{x} - \vec{x}' \in W_k \cap (W_1 + \dots + W_{k-1}) = \{\vec{0}\}$$

$$\Rightarrow \vec{x}_k = \vec{x}'_k \quad \text{ad} \quad \vec{x} = \vec{x}'$$

Similarly, one can uniquely decompose $\vec{x} = \vec{z}_{k-1} + \vec{z}$ with

$\vec{z}_{k-1} \in W_{k-1}$ ad $\vec{z} \in W_1 + \dots + W_{k-2}$ ad so on...

2). why not just use the condition $W_i \cap W_j = \{\vec{0}\}$.

Counter example: $W_1 = xy\text{-plane in } \mathbb{R}^3$

$W_2 = z\text{-axis}$

$W_3 = \text{any line through the origin but not } xy\text{-plane or } z\text{-axis.}$

Then $W_1 + W_2 + W_3$ is not a direct sum as one can see that $\forall \vec{z} \in \mathbb{R}^3$, there is more

than one decomposition of $\vec{z}$ into vectors on W_1, W_2 ad W_3 .

Proposition: Let $W_1, \dots, W_k \subseteq V$ be subspaces. If the sum $W_1 + W_2 + \dots + W_k$ is direct sum, Then $\dim(W_1 \oplus W_2 \oplus \dots \oplus W_k) = \dim W_1 + \dots + \dim W_k$

Proof Suppose $\dim W_i = d_i$ for all $i=1, \dots, k$

Let $B_i = \{\vec{v}_i^{(1)}, \dots, \vec{v}_{d_i}^{(i)}\}$ be a basis for W_i

Claim $B = \bigcup_{i=1}^k \{\vec{v}_i^{(1)}, \dots, \vec{v}_{d_i}^{(i)}\}$ is a basis for $W_1 \oplus \dots \oplus W_k$.

① By the definition of the sum of $W_1 + \dots + W_k$, clearly

$$\text{Span } B = W_1 + \dots + W_k.$$

② Suppose $\sum_{i=1}^k a_i^{(i)} \vec{v}_i^{(i)} + \dots + a_{d_i}^{(i)} \vec{v}_{d_i}^{(i)} = \vec{0}$

$$\text{Then } \sum_{i=1}^{k-1} a_i^{(i)} \vec{v}_i^{(i)} + \dots + a_{d_i}^{(i)} \vec{v}_{d_i}^{(i)} = -a_1^{(k)} \vec{v}_1^{(k)} - \dots - a_{d_k}^{(k)} \vec{v}_{d_k}^{(k)}$$

$$\text{LHS} \in W_1 + W_2 + \dots + W_{k-1} \quad \text{RHS} \in W_k$$

$$\Rightarrow \text{LHS} = \text{RHS} \in (W_1 + \dots + W_{k-1}) \cap W_k = \{\vec{0}\}$$

$$\Rightarrow \text{RHS} = \vec{0} \Leftrightarrow -a_1^{(k)} \vec{v}_1^{(k)} - \dots - a_{d_k}^{(k)} \vec{v}_{d_k}^{(k)}$$

As $B_k = \{ \vec{v}_1^{(k)}, \dots, \vec{v}_{d_k}^{(k)} \}$ is linearly independent

$$a_1^{(k)} = \dots = a_{d_k}^{(k)} = 0.$$

$$\Rightarrow \sum_{i=1}^{k-1} a_i^{(i)} \vec{v}_i^{(i)} + \dots + a_{d_i}^{(i)} \vec{v}_{d_i}^{(i)} = \vec{0}$$

Similarly, use the condition $W_{k-1} \cap (W_1 + W_2 + \dots + W_{k-2}) = \{\vec{0}\}$

$$\text{We may conclude } a_1^{(k-1)} = \dots = a_{d_{k-1}}^{(k-1)} = 0$$

Continue with similar arguments, we may prove all these $a_j^{(i)}$'s are zeros.

$\Rightarrow$ vectors in $\bigcup_{i=1}^k B_i$ are linearly independent.

Proposition. Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues. Then

$E_{\lambda_1} + E_{\lambda_2} + \dots + E_{\lambda_k}$ is a direct sum.

Proof. We need to prove $E_{\lambda_j} \cap E_{\lambda_{j-1}} + E_{\lambda_{j-2}} + \dots + E_{\lambda_2} + E_{\lambda_1} = \{\vec{0}\}$, $j=2, \dots, k$

$$\forall \vec{y} \in E_{\lambda_j} \cap E_{\lambda_{j-1}} + E_{\lambda_{j-2}} + \dots + E_{\lambda_2} + E_{\lambda_1}.$$

$$\exists \vec{x}_1 \in E_{\lambda_1}, \dots, \vec{x}_{j-1} \in E_{\lambda_{j-1}} \text{ such that } \vec{y} = \vec{x}_1 + \dots + \vec{x}_{j-1} \in E_{\lambda_j}$$

$$\Rightarrow \vec{x}_1 + \dots + \vec{x}_{j-1} - \vec{y} = \vec{0}$$

$$\text{Denote } \vec{x}_j = -\vec{y} \in E_{\lambda_j}$$

$$\text{Then } \vec{x}_1 + \dots + \vec{x}_{j-1} + \vec{x}_j = \vec{0}$$

$$\Rightarrow \vec{x}_1 = \dots = \vec{x}_j = \vec{0} \text{ by the corollary.}$$

$$\Rightarrow E_{\lambda_j} \cap E_{\lambda_{j-1}} + \dots + E_{\lambda_1} = \{\vec{0}\}$$

Definition: A linear transformation $T: V \rightarrow V$ is called diagonalizable if there exists an ordered basis B of V such that $[T]_B$ is a diagonal matrix.

Theorem. Let V be an n -dimensional vector space. Let $T: V \rightarrow V$ be a linear transformation. Let $\lambda_1, \dots, \lambda_k$ be all the distinct eigenvalues of T . Let B_i be an ordered basis for E_{λ_i} , for all $i=1, 2, \dots, k$. Then the following statements are equivalent.

1). T is diagonalizable

2). The characteristic polynomial of T is

$$f(\lambda) = (\lambda - \lambda_1)^{d_1} (\lambda - \lambda_2)^{d_2} \dots (\lambda - \lambda_k)^{d_k}$$

with $d_1 + \dots + d_k = n$ and $d_1 = \dim E_{\lambda_1}$, $d_2 = \dim E_{\lambda_2}$, $\dots$, $d_k = \dim E_{\lambda_k}$.

3). $\dim(E_{\lambda_1}) + \dim(E_{\lambda_2}) + \dots + \dim(E_{\lambda_k}) = n$

4). $B_1 \cup B_2 \cup \dots \cup B_k$ forms a basis for V .

(5). the minimal polynomial of T has no repeated roots.

end of Feb 10

Proof: (1) $\Rightarrow$ (2): Suppose T is diagonalizable.

$\Rightarrow$ there exists a basis $B = \{\vec{v}_1, \dots, \vec{v}_n\}$ such that

$$[T]_B = \begin{pmatrix} \alpha_1 & & \\ & \ddots & \\ & & \alpha_n \end{pmatrix}$$

$\Rightarrow f(\lambda) = \det(T - \lambda I) = (\alpha_1 - \lambda) \dots (\alpha_n - \lambda) \Rightarrow \alpha_1, \alpha_2, \dots, \alpha_n$ are all the eigenvalues